

Package ‘scScale’

June 27, 2026

Title Single-Cell Scaling Law Utilities

Version 0.0.1

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Description Median Marchenko-Pastur noise calibration, BBP spike inversion, spectral overlap, and task-difficulty utilities for single-cell scaling analyses, aligned with ChesleaZ/scScale.

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URL <https://github.com/ChesleaZ/scScale>

BugReports <https://github.com/ChesleaZ/scScale/issues>

Encoding UTF-8

Depends R (>= 2.10),
Matrix

LazyData true

LazyDataCompression xz

Suggests irlba,
SeuratObject

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

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gse123025_myeloid_hvg *GSE123025 Myeloid HVG Count Matrix Example*

Description

A larger example subset derived from GSE123025 for RMT spike-model tutorials. The object keeps all cells from the source matrix and restricts genes to 2,000 highly variable genes selected by `select_hvgs()`.

Usage

```
data(gse123025_myeloid_hvg)
```

Format

A list with:

counts Integer matrix with 2,000 selected genes/features by 1,922 cells.

source Source file name in the canonical lab data directory.

accession GEO accession identifier, "GSE123025".

description Short description of the HVG subset.

selection List describing the feature and cell selection.

original_dim Dimensions of the source matrix before HVG selection.

cell_totals Column sums of the HVG count matrix.

gene_totals Row sums of the HVG count matrix.

Source

Derived from `data/GSE123025/GSE123025_Single_myeloid_1922_cells_processed_data.csv.gz` in the canonical lab workspace. The source matrix has 23,429 genes by 1,922 cells and is about 8.0 MB compressed as CSV gzip, or 95 MB uncompressed.

Examples

```
data(gse123025_myeloid_hvg)
counts <- gse123025_myeloid_hvg$counts
dim(counts)
fit <- scscale_fit(counts, r = 10, mp_max_iter = 3, fit_umi = FALSE)
fit
```

`gse164378_3p_citeseq_hvg`*GSE164378 3 Prime CITE-seq RNA HVG and ADT Example*

Description

A compact paired RNA and ADT example derived from the 3 prime CITE-seq portion of GSE164378. The object keeps 4,000 matched PBMC cells, 2,000 RNA highly variable genes, and all 228 measured ADT features.

Usage

```
data(gse164378_3p_citeseq_hvg)
```

Format

A list with:

rna_counts Sparse integer matrix with 2,000 selected RNA genes by 4,000 cells.

adt_counts Sparse integer matrix with 228 ADT features by the same 4,000 cells.

meta Cell metadata, including RNA/ADT QC totals, lane, donor, time, batch, and cell type labels.

source Short source dataset description.

accession GEO accession identifier, "GSE164378".

description Short description of the compact paired-modality subset.

selection List describing the cell and RNA feature selection.

original_dim Dimensions of the source 3 prime RNA and ADT matrices before subsetting.

cell_totals RNA and ADT column sums for the selected cells.

feature_totals RNA and ADT row sums for the selected features.

Source

Derived from the 3 prime RNA and ADT matrices in GSE164378, the Hao et al. multimodal PBMC reference dataset. The source 3 prime matrices have 33,538 RNA features by 161,764 cells and 228 ADT features by the same 161,764 cells.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts
adt <- gse164378_3p_citeseq_hvg$adt_counts
dim(rna)
dim(adt)
```

Description

Object-style Gaussian spike fitting and spectral mutual information utilities for single-cell scaling-law analyses.

Usage

```
scscale_fit(x, input = c("counts", "normalized", "representation",
  "eigenvalues"), target = NULL, target_fit = NULL, n = NULL, p = NULL,
  target_depth = 1e4, center = TRUE, scale = TRUE, n_features = NULL,
  min_cells = 10, mp_max_iter = 50, mp_grid_n = 20000,
  mp_stop_metric = c("qq_rmse_log", "ks", "neg_log_likelihood", "none"),
  mp_stop_tol = 1e-4, mp_min_iter = 2, r = NULL,
  fit_umi = input[1] == "counts",
  sampling_rates = c(0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00),
  U_grid = NULL, umi_seed = 1, umi_replicates = 1,
  store_matrix = FALSE, use_irlba = TRUE)

scscale_mi(fit, target_fit, n_grid = NULL, U_grid = NULL,
  sampling_rates = NULL, combine = TRUE, r = NULL, eps = 1e-12)
```

Arguments

<code>x</code>	A feature-by-cell count matrix, sparse Matrix , normalized feature-by-cell matrix, low-dimensional representation, or covariance eigenvalue vector.
<code>input</code>	Input type. "counts" applies library-size normalization, log1p, and row standardization. "eigenvalues" requires <code>n</code> and <code>p</code> .
<code>target</code>	Optional empirical target.
<code>target_fit</code>	Optional target-side <code>scscale_fit</code> object.
<code>n, p</code>	Cell and feature counts, required when fitting directly from eigenvalues.
<code>target_depth, center, scale, n_features, min_cells</code>	Preprocessing settings for count matrices.
<code>mp_max_iter, mp_grid_n, mp_stop_metric, mp_stop_tol, mp_min_iter</code>	MP bulk fitting controls.
<code>r</code>	Number of spikes or singular vectors to keep.
<code>fit_umi</code>	Whether <code>scscale_fit()</code> should also estimate UMI scaling curves.
<code>sampling_rates, U_grid</code>	Sequencing-depth grid as global sampling rates or expected total UMI counts.
<code>umi_seed, umi_replicates</code>	Random seed and replicate count for stochastic UMI downsampling.
<code>store_matrix</code>	Whether to store the normalized matrix inside the returned fit object.
<code>use_irlba</code>	Whether to use irlba for partial singular-vector computations when available.
<code>fit</code>	An <code>scscale_fit</code> object for the measured modality.
<code>n_grid</code>	Optional cell-number grid for MI scaling.

combine	If TRUE, return the full cell-number by UMI grid when both grids are supplied to <code>scscale_mi()</code> .
eps	Numerical clipping value for MI calculations.

Details

The public workflow is:

1. Fit each modality with `scscale_fit()`.
2. Use one fitted object as the target in `scscale_mi()`.
3. Optionally provide cell-number and/or UMI grids to evaluate scaling laws.

Value

`scscale_fit()` returns an `scscale_fit` object with dimensions, normalization metadata, covariance spectrum, MP bulk fit, spike table, finite and infinite recoverability vectors, optional UMI scaling, and optional MI results. `scscale_mi()` returns theoretical MI at the fitted point and, when requested, UMI, cell-number, or full cell-by-UMI scaling tables.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[, 1:200]
adt <- gse164378_3p_citeseq_hvg$adt_counts[, 1:200]

fit_rna <- scscale_fit(rna, r = 3, fit_umi = FALSE, mp_grid_n = 500)
fit_adt <- scscale_fit(adt, r = 3, fit_umi = FALSE, mp_grid_n = 500)
scscale_mi(fit_rna, fit_adt, n_grid = c(100, 200), r = 3)$cell_scaling
```

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